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Table 4.9
Total Revenue - Price declining

Quantity sold (Q)	Price (P)	Total Revenue (TR)
1	10	10
2	9	18
3	8	24
4	7	28
5	6	30
6	5	30
7	4	28
8	3	24
9	2	18
10	1	10

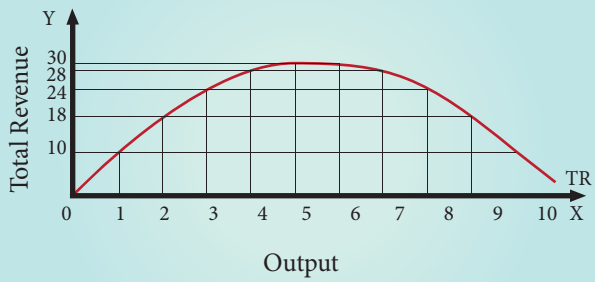


Diagram 4.11

TR = PQ = 1 × 10 = 10

When P = 3, Q = 8, TR = 24

When P = 0, Q = 1, TR = 10

b) Average Revenue

Average revenue is the revenue per unit of the commodity sold. It is calculated by dividing the Total Revenue(TR) by the number of units sold (Q)

AR = TR /Q; if TR = PQ, AR = PQ/Q = P

AR denotes Average Revenue, TR denotes Total Revenue and Q denotes Quantity of unit sold.

For example, if the Total Revenue from the sale of 5 units is Rs 30, the Average Revenue is Rs.6. **(AR= 30/5 =6)** It is to be noted that AR is equal to Price.

$$AR=TR/Q = PQ/Q=P$$

c) Marginal Revenue

Marginal revenue (MR) is the addition to the total revenue by the sale of an additional unit of a commodity. MR can be found out by dividing change in total revenue by the change in quantity sold out. MR = ΔTR / ΔQ where MR denotes Marginal Revenue, ΔTR denotes change in Total Revenue and ΔQ denotes change in total quantity.

The other method of estimating MR is:

$$MR=TR_n - TR_{n-1} \text{ (or) } TR_{n+1} - TR_n$$

where, MR denotes Marginal Revenue, TR_n denotes total revenue of nth item, TR_{n-1} denotes Total Revenue of $n\text{-}1^{th}$ item and TR_{n+1} denotes Total Revenue of $n+1^{th}$item.

If TR = PQ MR = dTR/dQ = P,

which is equal to AR.

4.6.2 Relationship between AR and MR Curves

If a firm is able to sell additional units at the same price then AR and MR will be constant and equal. If the firm is able to sell additional units only by reducing the price, then both AR and MR will fall and be different .

Constant AR and MR (at Fixed Price)

When price remains constant or fixed, the MR will be also constant and will coincide with AR. Under perfect competition as the price is uniform and fixed, AR is equal to MR and their shape will be a straight